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Glasgow College, UESTC

FINAL YEAR PROJECT REPORT
BACHELOR OF ENGINEERING

Information Compression Techniques for Control Data Exchange in Joint Communication and Control system

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2025-2026

Coursework Declaration and Feedback Form

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Course Code : UESTC4006P	Course Name : INDIVIDUAL PROJECT 4
Name of 1st Supervisor: Olaoluwa Popoola	Name of 2nd Supervisor:
Title of Project: Information Compression Techniques for Control Data Exchange in Joint Communication and Control system	
Declaration of Originality and Submission Information	
<i>I affirm that this submission is all my own work in accordance with the University of Glasgow Regulations and the School of Engineering requirements</i> Signed (Student) :	 UESTC4006P TEC-IT.COM
Date of Submission :	
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Abstract

The design of Joint Communication and Control (JCC) systems is critical for modern cyber-physical applications, where strict bandwidth constraints often lead to severe control instability and degraded performance. To address the fundamental challenge of maintaining control accuracy under extreme network limitations, this thesis investigates an intelligent, context-aware information compression strategy utilizing a multi-degree-of-freedom robotic manipulator as a comprehensive testbed. The proposed framework integrates an Event-Triggered Control (ETC) mechanism to reduce communication frequency in the temporal domain with a Deep Reinforcement Learning (DRL) agent—specifically utilizing Proximal Policy Optimization (PPO)—acting as a dynamic bandwidth allocator in the spatial domain. Unlike traditional uniform quantization or static allocation methods, the PPO agent learns a "pragmatic communication" strategy by observing real-time physical states and optimizing a reward function derived from the Linear Quadratic Regulator (LQR) tracking cost. This formulation enables the agent to dynamically distribute a strictly limited bit budget across different robotic joints, proactively sacrificing the precision of low-sensitivity linkages to protect highly sensitive base joints during critical movements. Simulation conducted on a PyBullet-based platform demonstrate the superiority of this co-design, which substantially reduces the overall communication payload—consuming less 6% of the ideal bandwidth—while effectively mitigating trajectory jittering and preserving near-ideal tracking precision, thereby validating the immense potential AI-driven dynamic resource allocation in enhancing the resilience of resource-constrained networked control systems.

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor

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1 Introduction

1.1 Background and Motivation

With the rapid advancement of wireless communication networks and distributed computing architectures, traditional isolated robotic systems are undergoing a profound paradigm shift toward Cloud Robotics and Teleoperation [6]. In these modern applications, the intensive computational and decision-making tasks multi-degree-of-freedom (Multi-DOF) robotic manipulators are increasingly offloaded to edge servers or cloud infrastructure. Such systems, where sensors, controllers, and actuators are interconnected via shared communication networks to form a cyber-physical closed loop, are defined as Networked Control Systems (NCS) [5]. NCS endows robotic systems with unprecedented flexibility, scalability, and collaborative computing capabilities, significantly reducing the hardware costs and local power consumption of individual robots.

However, incorporating communication networks into physical control loops introduces severe engineering and theoretical challenges. To maintain smooth and precise physical trajectory tracking, high-DOF robotic manipulators (e.g., 6-DOF and above) typically require extremely high-frequency state updates, often exceeding 1000 Hz in sensor sampling rates [12]. In stark contrast, real-world wireless communication channels are subject to strict bandwidth constraints and exhibit highly dynamic, unpredictable behaviors. When sensors inject massive volumes of full-precision state data into the network at high frequencies, the limited channel capacity is rapidly exhausted, triggering severe network congestion, packet loss, and unpredictable network-induced delays.

Under extreme network resource constraints, the traditional "Best-Effort" data transmission protocols are fundamentally inadequate for high-precision physical control [4]. Deteriorating communication quality directly disrupts the timeliness and integrity of the control feedback loop, causing severe trajectory jittering, control divergence, and ultimately, system failure during delicate manipulation tasks. Consequently, the critical challenge in modern Cyber-Physical Systems (CPS) lies in drastically reducing the communication data payload under extreme bandwidth limitations while simultaneously guaranteeing the control stability and tracking precision of the robot.

1.2 The Fallacy of Separation and the Need for JCC

Historically, the design of networked control systems has been heavily influenced by Shannons Separation Principle, which posits that source coding (data compression) and channel coding (error protection) can be designed independently without any loss of optimality. In the context of robotic control, this principle translates into a rigid decoupling between communication protocols and control algorithms [11]. Communication engineers typically design systems to minimize the generic reconstruction error of the transmitted data often utilizing uniform quantization and periodic transmission mechanisms while control engineers develop algorithms under the unrealistic assumption of perfect, continuous data reception. However, this artificial separation is fundamentally flawed in resource-constrained cyber-physical systems. Treating all sensory data equally blindly ignores the intrinsic "control-criticality" of the physical states. In multi-DOF robotic manipulators, minor deviations in specific joints can lead to drastically different physical consequences, rendering uniform compression highly inefficient and prone to causing catastrophic system instability [7].

To overcome the limitations of the separation paradigm, this thesis advocates for a Joint Communication and Control (JCC) framework. The core philosophy of JCC is that when com-

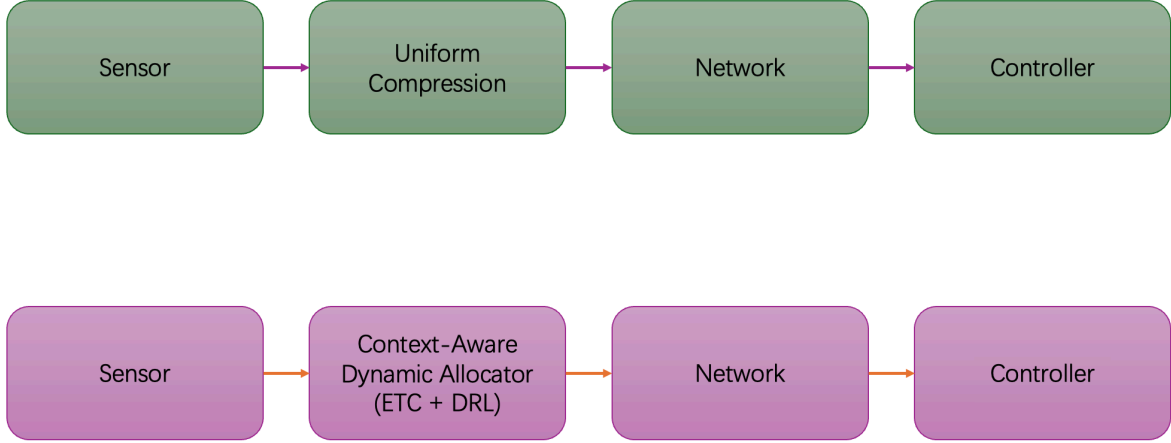


Figure 1: System Paradigm Comparison between Traditional NCS and Proposed JCC.

munication bandwidth is severely restricted, data transmission and compression algorithms must be explicitly control-oriented and context-aware. Instead of blindly minimizing the mathematical mean square error (MSE) of the data stream, the communication protocol must priority the transmission of information that is most vital to maintaining the stability and tracking precision of the physical plant.

Specifically, this thesis implements the JCC paradigm through two complementary technical approaches. In the temporal domain, the rigid, high-frequency periodic transmission is replaced by an Event-Triggered Control (ETC) mechanism [10]. ETC ensures that sensor data is transmitted over the network only when the physical state deviation crosses a predefined safety threshold, effectively eliminating redundant data exchange during stable operational phases. In the spatial domain, the proposed framework exploits the mechanical lever effect inherent in robotic manipulators—where errors in base joints exponentially magnify end-effector deviations compared to errors in distal joints. By dynamically evaluating the real-time sensitivity of each joint, the system executes non-uniform, weighted bandwidth allocation, stripping communication resources from robust linkages to guarantee the high-precision transmission of critical joint states [2].

1.3 AI-Driven Dynamic Allocation

While static, control-aware allocation methods such as those based on Linear Quadratic Regulator (LQR) sensitivity matrices offer significant improvements over uniform compression, they suffer from inherent limitations. Previous research in NCS has extensively explored static resource allocation and Value-of-Information (VoI) based scheduling to optimize bandwidth usage [3]. However, these traditional methodologies largely rely on linearized models and fixed sensitivity weightings. In real-world applications, the dynamics of multi-DOF robotic manipulators are highly nonlinear; the sensitivity of each joint varies dynamically in real-time depending on the robot’s current posture, velocity, and external payloads. Consequently, static allocation policies inevitably result in sub-optimal performance during complex, high-speed maneuvers, driving the need for task-oriented or goal-oriented communication frameworks [8].

To overcome the limitations of static models and resolve this complex, nonlinear dynamic resource allocation game, this thesis introduces Deep Reinforcement Learning (DRL) into the communication protocol design. Recently, DRL has demonstrated exceptional capabilities in

handling high-dimensional, continuous control and scheduling problems within cyber-physical environments, bridging the gap between optimal control theory and adaptive communication [1]. Specifically, this work employs the Proximal Policy Optimization (PPO) algorithm [9], which is highly favored for its robust convergence properties, sample efficiency, and stability in continuous action spaces.

Driven by the PPO algorithm, the framework realizes the philosophy of "Pragmatic Communication". Instead of following predefined mathematical rules, the AI agent acts as a centralized bandwidth arbiter. It continuously observes the real-time physical tracking errors and environmental states. By optimizing a reward function strictly tied to the minimization of the overall control cost, the agent organically learns a pragmatic strategy. During critical maneuvers, the DRL agent proactively strips communication bandwidth from low-sensitivity linkages and re-allocates the strictly limited bit budget to protect highly sensitive base joints. This learning-based dynamic allocation ensures optimal system resilience under extreme network constraints.

1.4 Research Objectives and Contributions

The primary objective of this thesis is to formulate, implement, and evaluate an intelligent, Deep Reinforcement Learning (DRL)-driven Joint Communication and Control (JCC) framework. This framework aims to guarantee precise and stable trajectory tracking for multi-degree-of-freedom (Multi-DOF) robotic manipulators operating over extreme bottleneck communication channels specifically under conditions where available bandwidth is severely restricted to less than 10% of the ideal requirement. By seamlessly integrating temporal event-triggered mechanisms with spatial dynamic bandwidth allocation, this research seeks to resolve the fundamental trade-off between communication efficiency and physical control stability in modern networked cyber-physical systems.

To achieve the aforementioned objectives, the core contributions of this thesis are summarized as follows:

1. **Development of a High-Fidelity Cyber-Physical Testbed:** A comprehensive closed-loop control simulation environment was constructed utilizing the PyBullet physics engine. This testbed faithfully replicates realistic kinematic and dynamic constraints of robotic manipulators, providing a robust and reproducible foundation for evaluating networked control algorithms.
2. **Implementation of a Hybrid ETC and LQR-Weighted Benchmark:** A baseline evaluation system was designed by integrating Event-Triggered Control (ETC) in the temporal domain with Linear Quadratic Regulator (LQR)-based static weighted compression in the spatial domain. This benchmark successfully validates the necessity of control-aware, non-uniform data compression over traditional periodic and uniform methods.
3. **Design of a Novel DRL-Driven Context-Aware Allocator:** As the primary innovation, this thesis proposes a dynamic bandwidth allocator powered by the Proximal Policy Optimization (PPO) algorithm. The DRL agent organically learns a "pragmatic communication" strategy by directly observing nonlinear tracking errors and optimizing real-time bandwidth distribution without relying on fixed linear mathematical approximations.
4. **Empirical Validation of Extreme Bandwidth Compression:** Extensive experimental evaluations demonstrate the superiority of the proposed PPO-driven framework. The intelligent agent successfully compressed the communication data payload to under 6%

of the ideal baseline channel capacity, while effectively eliminating trajectory jitter and maintaining near-perfect tracking precision for highly sensitive joints.

1.5 Thesis Organization

2 Body of the report

2.1 Structure and sections

The structure of the report and the titles of sections depend on the nature of the project. Here are suggestions for two extreme types, which can be adjusted to suit your report.

2.1.1 Research projects

Here you are given a starting point and a general direction: your aim is to discover something new. The report would probably be divided into sections with familiar titles.

- **Theory** – Explain essential background theory that is vital to understand your report. Select and present the material to show its relevance to the project and demonstrate your understanding. Do not repeat standard material from textbooks, which is a waste of space; give a reference instead.
- **Experimental Techniques** – Give an account of your experimental (or numerical) techniques. Emphasise details that would be necessary to continue the work or to compare your results with those of other experimenters. Aim to include what you would have liked to know at the start of your project.
- **Results** – Describe these at an appropriate level of detail to support your conclusions. Lengthy tables should be left to an appendix or supplied on a CD. Avoid unnecessary detail: describe preliminary experiments briefly if at all, unless they illustrate some important point.

2.1.2 Design, build and test projects

Here the goal is specified, more or less precisely: your job is to work out how to get there. The titles of the sections are less standardised but here is a possible approach.

- **Specification** – Typically you are given only a vague description of the functions required and must first develop this into a firm specification against which your final product will be judged.
- **Possible strategies** – Evaluate *briefly* the possible approaches that you considered and explain why you selected one of these. Avoid excessive detail of discarded possibilities.
- **Implementation** – Describe the final product. Concentrate on key features that required advanced design and skim over well-known aspects. Mention useful points that might help future students, such as tricky sections of data sheets.

Software is tricky. Often the best approach is to describe the high-level structure in the main text (a diagram of a state machine for example) and pick out any special features of the program. A complete listing should be included as an appendix or on a CD.

- **Testing** – This is equivalent to the ‘Results’ section of a research project.

2.2 Style of writing

One of the most difficult aspects of writing is to judge the level and background of your readers. Typically the report will be assessed by your first supervisor, who is an expert on the topic, and your second supervisor, who is not. You should therefore assume that the reader is a *well-educated, graduate engineer* but not an expert on the subject of the project. Assume that he or she is familiar with the general concepts taught in courses up to level 3 but not with the details of specialised courses at higher levels.

It is also difficult to appreciate that most readers are not interested in the nitty-gritty detail. They want to know *what* you did and *why* you did it that way but they don't want a step-by-step account of *how* you did it. Your watchword should therefore be:

SUMMARISE!

Of course some people need the details – a person continuing work on the project requires precise experimental methods, for instance. Such material may be better in an appendix. The same is true for computer programs and schematics of complicated circuits. Design-and-build projects may need a User's Manual, which should be included as an appendix.

A report is a formal document and should be written in appropriate language. Numerous books offer advice on writing reports and a selection [jve05, mkm06, dfb09, tmy05] is listed in the references at the end. Here are a few tips.

- Reports should be written in correct English. Break text into paragraphs, keep sentences to a reasonable length and insert appropriate punctuation. Use a spell-checker and a grammar-checker if desired but neither is a substitute for careful reading.
- A report is not a story. Write 'The voltage was measured' rather than 'I measured the voltage'. This document contains instructions and therefore uses a different style.
- Define all abbreviations when they are first used: 'The accelerometer uses a serial peripheral interface (SPI)'. Provide a list of abbreviations if you use a large number of them.
- Don't write material that you don't understand. It will be obvious to the reader.

The quality of English is assessed as part of the report. Foreign students may feel this to be a burden but part of their education in this country is to learn to work effectively in an English-speaking environment.

2.3 Precision

An engineering report must be precise. This applies both to the language and to numerical values. For example, the words *precision* and *accuracy* are often used interchangeably in non-technical discussion but the distinction between them is vital in engineering.

Quote numerical results to an appropriate number of significant figures. For example, it is pointless to claim that a length was 12.345 mm if it was measured with a pocket ruler. Don't simply write down all the digits displayed on your calculator.

Analyse the uncertainties in your results to increase the impact of your results. Please avoid horrors like this:

The gain was quite accurate.

The sentence is meaningless and the reader will doubt whether you have any idea of the accuracy. Contrast this sentence:

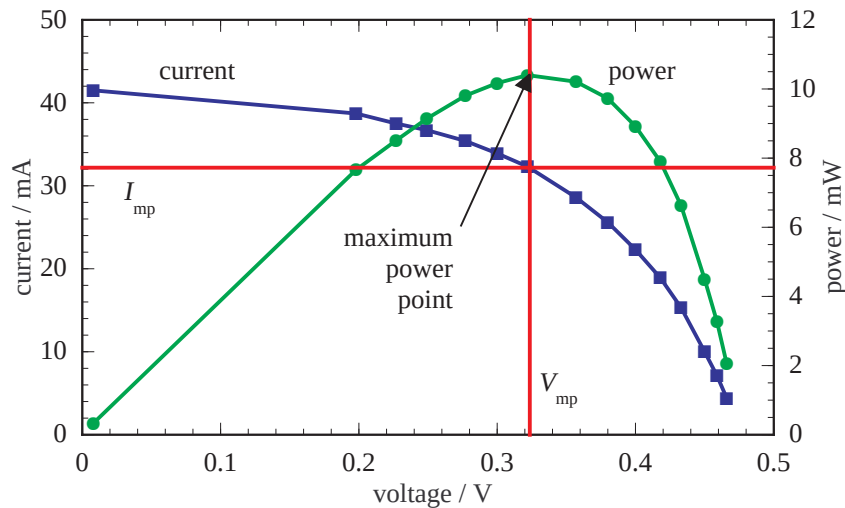


Figure 2: Current and power as a function of voltage delivered by a single photovoltaic cell under illumination from a 150 W metal halide floodlamp, showing the maximum power point.

The accuracy of the gain was estimated to be $\pm 2\%$, limited by the tolerance of the resistors. A detailed analysis is given in Appendix C.

This is informative and convinces the reader that you have a full understanding.

2.4 Figures and tables

Figures and tables must have informative captions and be numbered as in Figure 2. Axes of graphs should have scales, titles and units, otherwise the plot is meaningless. All text must be legible, roughly the same size as the main text. Be warned that plots from Excel need extensive editing to bring them up to an acceptable standard. Oscilloscope screenshots can make good illustrations and are simple to capture on modern equipment.

Tables must have appropriate headings with units. Avoid lengthy tables: consider whether a graph would be clearer or the data would be better on a CD.

2.5 References

No project is done in isolation. A research project builds on the results of previous workers; a design-and-build project depends on the properties of the components available. You therefore draw on published documents during your project and must provide *references* to these sources in your report. References are cited (a) to give due credit to the originator and (b) to guide a reader who wants more detailed information. You should give a reference wherever it is required for either of these purposes. Properly referenced material from many sources is a sign of a good project report.

References must be cited with sequential numbers in square brackets where they are used in the text, not just listed at the end of the report. Typical usage is ‘Fitzmaurice and Hand [gmf87] showed that...’ or ‘The median is an appropriate estimator for this signal [gmf87]’.

Avoid direct quotations from references in general; make it absolutely clear that the text is a quotation if this is unavoidable. An illustration, such as a diagram from a data sheet, is another type of quotation and must be referenced, typically in the figure caption. This is perfectly acceptable: do not waste time redrawing diagrams.

2.5.1 Plagiarism

If you copy from another person's work (project report, book, journal, web page or any other document) without acknowledging the source, you are guilty of *plagiarism*. This is a disciplinary offence and the University has procedures for handling it. The failure to acknowledge a source is considered as plagiarism even if there was no deliberate intention to cheat.

Avoid any risk of plagiarism by providing a reference for all sources that you use. Read the advice offered by the University Library and consult your supervisor if you are in any doubt. The University's *Plagiarism Statement* is reproduced in Appendix B.

2.5.2 List of references

All reports must have a section entitled 'References' after the main text but before any appendices. This comprises a list of references, numbered to match the citations in the text. Each reference requires the following information and the cited references provide examples.

- **Journal paper:** author(s), title of paper, name of journal, volume number, first page, last page and year [gmf87]. Many journals now use article numbers instead of page numbers.
- **Book:** author(s), title, edition, publisher and year of publication [jve05]. Include the number of the chapter or page(s) if you refer to only a small part of the book.
- **Data sheet:** company, title, edition and date [s08qg]. Application notes, technical reports and similar documents should be cited in the same way.
- **Web page:** author(s) or organisation, title, full URL and date of viewing [mac09]. See below.

The reader must be able to find the document without searching for further information.

2.5.3 References from the Web

The World Wide Web is a wonderful resource because it is so easy to search. It is therefore tempting to use web pages as references. *Proceed with caution* because the accuracy of many web sites cannot be verified. This is particularly true for anonymous sites such as Wikipedia. Use them only as a starting point: good pages provide references to more authoritative sources.

Reports whose references are all or mainly from the web, especially from anonymous sites, will be penalised.

3 Discussion

The discussion is your opportunity to impress an expert with your depth of understanding. Don't worry so much about the non-expert reader – she or he can skip to the Conclusions.

For a research project this section should provide a logical argument leading from the experimental observations to the final conclusions. Evaluate your results thoroughly and gain every possible piece of understanding from them. Compare the results in detail to other work in the field. This should be a *critical* comparison so don't just say that your results are different from previous work; explain *why*. Never say 'The results were as expected' unless you have already described exactly what was expected!

The main purpose of the Discussion in a design-and-build project is to assess the performance of your product against the specification, showing its strengths and weaknesses.

4 Conclusions

Every report must have conclusions, built on the discussion. This section includes a summary of the main achievements of the work but is more than that. The noun ‘conclusion’ can be defined as ‘a judgement or decision reached by reasoning’ [**oupconc**] so you should highlight what has been learnt as a result of your project. What is the big picture that the reader should take away?

The Conclusions should include an assessment of the outcome against the original aims of the project. If you were unable to fulfil some aims, explain why. It is also useful to review the project plan (which should be included with the report). Did some tasks prove to be unexpectedly difficult, for instance?

4.1 Suggestions for further work

Almost every project leaves you with ideas for the future. Research typically answers some questions while opening new ones; by the end of a design-and-build project you may have thought of a superior approach. Examiners are impressed by intelligent suggestions for further work because they show that you really understand the project. It is often a sign of strength to identify the weak points and suggest solutions for them.

References

- [1] Marun K. Mitra, *Effective Technical Communication: A Guide for Scientists and Engineers*, Oxford (2006).
- [2] David F. Beer and David McMurrey, *A Guide to Writing as an Engineer*, 3rd Edition, Wiley (2009).
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- [5] Freescale Semiconductor, *Datasheet for MC9S08QG microcontroller*, revision 4 (2008).
- [6] Morgan Advanced Ceramics, *Advanced Ceramics: Silicon Nitride*, www.morgantechnicalceramics.com (viewed 2009 May 13).
- [7] Oxford University Press, *AskOxford*, www.askoxford.com/?view=uk (viewed 2009 May 13).

‘Conclusion’ is defined as (1) an end or finish, (2) the summing-up of an argument or text, (3) a judgement or decision reached by reasoning or (4) the settling of a treaty or agreement.

A Appendices

Use appendices for supporting material which is relevant but of a detailed nature. In this way the flow of ideas in the body of the report is not interrupted. Appendices are usually lettered rather than numbered to distinguish them. Several uses for appendices have been suggested already but bear in mind that a CD may be more appropriate for large tables of results, long programs and data sheets or other background information. Appendices are not included in the word count because the assessors are not obliged to read them. Do not put vital points here!

Include your project plan and interim reports as appendices on paper.

B University's Plagiarism Statement

The University's degrees and other academic awards are given in recognition of a student's personal achievement. All work submitted by students for assessment is accepted on the understanding that it is the student's own effort.

Plagiarism is defined as the submission or presentation of work, in any form, which is not one's own, without acknowledgement of the sources. Plagiarism includes inappropriate collaboration with others. Special cases of plagiarism can arise from a student using his or her own previous work (termed auto-plagiarism or self-plagiarism). Autoplagerism includes using work that has already been submitted for assessment at this University or for any other academic award.

The incorporation of material without formal and proper acknowledgement (even with no deliberate intent to cheat) can constitute plagiarism. Work may be considered to be plagiarised if it consists of:

- a direct quotation;
- a close paraphrase;
- an unacknowledged summary of a source;
- direct copying or transcription.

With regard to essays, reports and dissertations, the rule is: if information or ideas are obtained from any source, that source must be acknowledged according to the appropriate convention in that discipline; and any direct quotation must be placed in quotation marks and the source cited immediately. Any failure to acknowledge adequately or to cite properly other sources in submitted work is plagiarism. Under examination conditions, material learnt by rote or close paraphrase will be expected to follow the usual rules of reference citation otherwise it will be considered as plagiarism. Departments should provide guidance on other appropriate use of references in examination conditions.

Plagiarism is considered to be an act of fraudulence and an offence against University discipline. Alleged plagiarism, at whatever stage of a student's studies, whether before or after graduation, will be investigated and dealt with appropriately by the University.

The University reserves the right to use plagiarism detection systems, which may be externally based, in the interests of improving academic standards when assessing student work.